10.27 Fermi Statistics and Fermion Signs

From LecNote16 on, spinor fields anticommute, and from LecNote17 p. 14 on, minus signs appear that scalar fields never had. A page number without a LecNote refers to the last LecNote named before it. Students stop at four places:

- "Without any surprise, they are fermions" (LecNote16 p. 7). Could I quantize Ψ with commutators, like φ? And how is the positive $Q=\int d^3x\,\Psi^\dagger\Psi$ the number of b-particles *minus* the number of d-particles (LecNote18 p. 9)?
- Why must the sources $\eta,\bar\eta$ anticommute, and why is there no $\frac{1}{2}$ in $Z_0(\bar\eta,\eta)=\exp[i\int\bar\eta S\eta]$ (LecNote17 pp. 14–15)?
- How do the two Wightman functions of LecNote19 p. 4, one with $-\not p+m$ and one with $-\not p-m$, make the single covariant S of p. 3?
- Where does the circled minus of LecNote18 p. 2 ("exchange of fermions") come from, and are there other signs?

In short: the v-modes carry negative frequency, and only anticommutators keep the energy bounded below. Anticommuting fields need Grassmann variables and sources, whose Gaussian integral is $\det M$, not $1/\det M$. The sign that T carries for fermions turns the two spin sums into S and produces the minus of exchange diagrams and a -1 per closed fermion loop.

Conventions are Srednicki's. The metric is $(-,+,+,+)$, $\{\gamma^\mu,\gamma^\nu\}=-2g^{\mu\nu}$, $px=\mathbf{p}\cdot\mathbf{x}-\omega t$ and $\widetilde{dp}=d^3p/\big((2\pi)^3 2\omega\big)$; $\bar\Psi=\Psi^\dagger\beta$ with β numerically equal to γ^0, and $(\gamma^0)^2=1$, so $\bar\Psi\gamma^0=\Psi^\dagger$; $\langle X\rangle$ means $\langle 0|X|0\rangle$. The free field has $\mathcal{L}_0=i\bar\Psi\,\partial\!\!\!/\Psi-m\bar\Psi\Psi$ and

$$\Psi(x)=\sum_{s=\pm}\int\widetilde{dp}\,\big[b_s(\mathbf{p})u_s(\mathbf{p})e^{ipx}+d_s^\dagger(\mathbf{p})v_s(\mathbf{p})e^{-ipx}\big],\tag{10.27.1}$$

with $(\not p+m)u_s=0$ and $(-\not p+m)v_s=0$. The spinor facts used below are derived in §10.26: the spin sums $\sum_s u_s\bar u_s=-\not p+m$ and $\sum_s v_s\bar v_s=-\not p-m$; the normalization $\bar u_s\gamma^0 u_{s'}=\bar v_s\gamma^0 v_{s'}=2\omega\delta_{ss'}$ at equal momenta; and $\bar u_s(\mathbf{p})\gamma^0 v_{s'}(-\mathbf{p})=\bar v_s(\mathbf{p})\gamma^0 u_{s'}(-\mathbf{p})=0$ (LecNote18 p. 9). The propagators are $\Delta(x)=\int\frac{d^4p}{(2\pi)^4}\,e^{ipx}/(p^2+m^2-i\epsilon)$ and $S=(i\,\partial\!\!\!/+m)\Delta$, with $(-\partial^2+m^2)\Delta=(-i\,\partial\!\!\!/+m)S=\delta^4$.

Commutators would leave the energy unbounded

By the Dirac equation, $(-i\gamma^i\partial_i+m)$ acting on $u\,e^{ipx}$ gives $+\omega\gamma^0 u\,e^{ipx}$, and acting on $v\,e^{-ipx}$ it gives $-\omega\gamma^0 v\,e^{-ipx}$: the v-modes carry negative frequency.

Insert (10.27.1) into $H = \int d^3x\, \bar\Psi(-i\gamma^i\partial_i + m)\Psi$ (LecNote18 p. 13; derived below). The terms with opposite momenta drop out by $\bar u(\mathbf{p})\gamma^0 v(-\mathbf{p}) = 0$, the others give 2ω, and, keeping each product of operators in the order in which it arrives,

$$H = \sum_s \int \widetilde{dp}\;\omega\,[b_s^\dagger b_s - d_s d_s^\dagger], \qquad Q = \sum_s \int \widetilde{dp}\;[b_s^\dagger b_s + d_s d_s^\dagger], \tag{10.27.2}$$

where $Q = \int d^3x\,\bar\Psi\gamma^0\Psi$ is the charge of $\Psi \to e^{-i\alpha}\Psi$ (LecNote17 p. 6). Nothing about b and d has been assumed yet. With $c \equiv (2\pi)^3\,2\omega\,\delta^3(\mathbf{0})$ and $\int$ standing for $\sum_s \int \widetilde{dp}$, the two choices give

$$\begin{aligned}
\{d, d^\dagger\} = c: &\quad H = \int \omega\,[b^\dagger b + d^\dagger d] - 4E_0V, \quad Q = \int [b^\dagger b - d^\dagger d] + \text{const},\\
[d, d^\dagger] = c: &\quad H = \int \omega\,[b^\dagger b - d^\dagger d] - 4E_0V, \quad Q = \int [b^\dagger b + d^\dagger d] + \text{const}.
\end{aligned} \tag{10.27.3}$$

With commutators each d-quantum lowers the energy by ω, so there is no ground state, and antiparticles carry the charge of particles. Calling $\tilde d = d^\dagger$ the annihilator does not help: $[\tilde d, \tilde d^\dagger] = -c$, and $\tilde d^\dagger|0\rangle$ has negative norm. With anticommutators H is bounded below, $b|0\rangle = d|0\rangle = 0$, and $Q = N_b - N_d$. The constant: per mode $\omega\, b^\dagger b - \omega\, d d^\dagger = \omega(N_b - \frac12) + \omega(N_d - \frac12)$, so each of the four oscillators per momentum (b and d, two spins) has zero-point energy $-\frac12\omega$, a boson's $+\frac12\omega$ reversed. They give $-\sum_s \int \widetilde{dp}\,\omega c = -2V\int \frac{d^3p}{(2\pi)^3}\,\omega = -4E_0V$, since $(2\pi)^3\delta^3(\mathbf{0}) = V$; E_0 is the zero-point energy density of one real scalar (LecNote18 p. 13's "Recall SHO?").

This is spin–statistics for the free Dirac field (Srednicki §39): fields of half-integer spin, which change sign under a 2π rotation (§10.25), are quantized with anticommutators. Causality, below, gives the same verdict.

Quantizing the Dirac field

Since $\bar\Psi\gamma^0 = \Psi^\dagger$, $\mathcal{L}_0 = i\Psi^\dagger\partial_0\Psi + \bar\Psi(i\gamma^i\partial_i - m)\Psi$ is first order in time: the momentum conjugate to Ψ is $\pi = i\Psi^\dagger$, not a second coordinate, and $H = \pi\partial_0\Psi - \mathcal{L}_0 = \bar\Psi(-i\gamma^i\partial_i + m)\Psi$. The canonical relations $\{\Psi_\alpha, \Psi_\beta\} = 0$ and $\{\Psi_\alpha(\mathbf{x},t), \pi_\beta(\mathbf{y},t)\} = i\delta_{\alpha\beta}\delta^3(\mathbf{x}-\mathbf{y})$ read

$$\{\Psi_\alpha(\mathbf{x},t), \bar\Psi_\beta(\mathbf{y},t)\} = (\gamma^0)_{\alpha\beta}\,\delta^3(\mathbf{x}-\mathbf{y}) \tag{10.27.4}$$

(LecNote18 p. 10; the γ^0 is the β of $\bar\Psi$). They are equivalent to

$$\{b_s(\mathbf{p}), b_{s'}^\dagger(\mathbf{p}')\} = \{d_s(\mathbf{p}), d_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3\,2\omega\,\delta^3(\mathbf{p}-\mathbf{p}')\,\delta_{ss'}, \tag{10.27.5}$$

all other anticommutators zero. Inserting (10.27.1),

$$\{\Psi(\mathbf{x},t),\bar\Psi(\mathbf{y},t)\}=\int \widetilde{dp}\, e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\sum_s\big[u_s\bar u_s(\mathbf{p})+v_s\bar v_s(-\mathbf{p})\big],$$

and the spin sums give $(-\not p+m)+(-\not p'-m)=2\omega\gamma^0$ with $p'=(\omega,-\mathbf{p})$: that is $\gamma^0\delta^3$. Conversely, projecting b_s and $d_s^\dagger$ out of Ψ with $\bar u_s\gamma^0$ and $\bar v_s\gamma^0$ (LecNote18 pp. 8–13) recovers (10.27.5) from (10.27.4).

Grassmann integrals and $Z_0(\bar\eta,\eta)$

Anticommuting operators need anticommuting path-integral variables (LecNote17 pp. 16–21, Srednicki §44; skipped in class). Both build the calculus the same way. Grassmann numbers anticommute, so $\theta_i^2=0$ and $f(\theta)=a+\theta b$ (his (44.3)–(44.4)); linearity and invariance under a shift by an odd constant force $\int d\theta=0$, and with $\int d\theta\,\theta=1$ ((44.9)) integration acts like differentiation (p. 17). Under $\theta_i=J_{ij}\theta'_j$, $d^n\theta=(\det J)^{-1}d^n\theta'$, the inverse of the commuting rule ((44.18)–(44.19); p. 18), and $\int d\chi\,d\bar\chi\,\bar\chi\chi=1$ ((44.31); p. 19). Diagonalizing M (p. 20) and shifting $\chi\to\chi-M^{-1}\eta$, $\bar\chi\to\bar\chi-\bar\eta M^{-1}$ (p. 21) give his (44.40),

$$\int\prod_i d\chi_i\,d\bar\chi_i\;\exp\big(\bar\chi M\chi+\bar\eta\chi+\bar\chi\eta\big)=\det M\;\exp\big(-\bar\eta M^{-1}\eta\big),\tag{10.27.6}$$

with his pairwise measure (44.35). With p. 17's order applied to each set, $d\chi_n\cdots d\chi_1\,d\bar\chi_n\cdots d\bar\chi_1$, the result is $(-1)^{n(n-1)/2}\det M$; the η dependence is unchanged. Commuting variables give $(2\pi)^n(\det M)^{-1}$ instead (p. 20's "can you prove it?"): with $z=(x+iy)/\sqrt2$, a unitary that diagonalizes a Hermitian $M>0$ has unit Jacobian, and each eigenvalue gives $\int dx\,dy\,e^{-\lambda(x^2+y^2)/2}=2\pi/\lambda$.

With $\mathcal{L}_0=-\bar\Psi D\Psi$, $D=-i\not\partial+m$, the exponent $i\int(-\bar\Psi D\Psi+\bar\eta\Psi+\bar\Psi\eta)$ of LecNote17 p. 15 is (10.27.6) with $M=-iD$, $\eta\to i\eta$ and $\bar\eta\to i\bar\eta$:

$$Z_0(\bar\eta,\eta)=\exp\Big[i\int d^4x\,d^4y\;\bar\eta(x)S(x-y)\eta(y)\Big],\qquad S=D^{-1},\tag{10.27.7}$$

once $Z_0(0,0)=1$ removes $\det(-iD)$. In momentum space $\tilde S=(\not p+m)^{-1}$, since $(\not p+m)(-\not p+m)=p^2+m^2$, and $S=(i\not\partial+m)\Delta$ inherits the $-i\epsilon$ of Δ (§10.8). There is no $\frac12$ because Ψ and $\bar\Psi$ are independent, each with its own source, like φ and $\varphi^\dagger$ in the complex scalar's $\exp[i\int J^\dagger\Delta J]$ (p. 14). The real scalar's $\frac12$ belongs to the quadratic form $\frac12\varphi K\varphi$ of a single field.

The sources are odd because $\bar\eta\Psi$ sits in the exponent beside $\mathcal{L}$, an ordinary commuting quantity, and Ψ is odd. Derivatives with respect to odd sources anticommute and pass odd factors with a sign:

$\frac{\delta}{\delta\eta}\int(\bar\eta\Psi+\bar\Psi\eta)=-\bar\Psi$, but $\frac{\delta}{\delta\bar\eta}$ of the same is $+\Psi$ (p. 15). Hence, with (10.27.7),

$$\Psi_\alpha(x)\leftrightarrow\frac{1}{i}\frac{\delta}{\delta\bar\eta_\alpha(x)},\qquad \bar\Psi_\beta(y)\leftrightarrow i\frac{\delta}{\delta\eta_\beta(y)},\qquad \langle T\Psi_\alpha(x)\bar\Psi_\beta(y)\rangle=\frac{1}{i}S_{\alpha\beta}(x-y). \tag{10.27.8}$$

Reversing two derivatives flips the sign, so correlators are antisymmetric under the exchange of two fermion fields: $\langle T\bar\Psi_\beta(y)\Psi_\alpha(x)\rangle=-\frac{1}{i}S_{\alpha\beta}(x-y)$. So the vertex $g\varphi\bar\Psi\Psi$ needs $i\frac{\delta}{\delta\eta_\alpha}$ to the left of $\frac{1}{i}\frac{\delta}{\delta\bar\eta_\alpha}$; LecNote17 p. 22's order gives $-g$.

The time-ordered product and the propagator

The operator route to S (LecNote19 pp. 3–4, Srednicki §42) uses the time-ordered product with the sign of the permutation (p. 2: "extra minus sign for fermions"),

$$T\Psi_\alpha(x)\bar\Psi_\beta(y)=\theta(x^0-y^0)\,\Psi_\alpha(x)\bar\Psi_\beta(y)-\theta(y^0-x^0)\,\bar\Psi_\beta(y)\Psi_\alpha(x), \tag{10.27.9}$$

the order that the anticommuting derivatives of (10.27.8) produce. With (10.27.1), (10.27.5) and the spin sums, the Wightman functions of p. 4 are

$$\begin{aligned}\langle\Psi(x)\bar\Psi(y)\rangle&=(i\,\not\partial_x+m)\,D_+(x-y),\\ \langle\bar\Psi(y)\Psi(x)\rangle&=-(i\,\not\partial_x+m)\,D_-(x-y),\end{aligned} \tag{10.27.10}$$

with $D_\pm(\xi)=\int\widetilde{dp}\,e^{\pm ip\xi}$; $D_+(x-y)$ is $\langle\varphi(x)\varphi(y)\rangle$ for a free scalar of mass m (not the Yukawa scalar, whose mass is M). The minus of $\sum_s v_s\bar v_s=-\not p-m$ stands in front of the second, and the minus of (10.27.9) cancels it:

$$\langle T\Psi(x)\bar\Psi(y)\rangle=(i\,\not\partial_x+m)\big[\theta(x^0-y^0)D_+ +\theta(y^0-x^0)D_-\big]-i\gamma^0\delta(x^0-y^0)\big[D_+-D_-\big].$$

The last term, from ∂_0 on the θ's, vanishes since $D_+=D_-$ at equal times, and the bracket is the scalar $\langle T\varphi(x)\varphi(y)\rangle=\frac{1}{i}\Delta(x-y)$. Hence

$$S(x-y)=i\langle T\Psi(x)\bar\Psi(y)\rangle=(i\,\not\partial_x+m)\,\Delta(x-y), \tag{10.27.11}$$

the S of (10.27.8), with the $-i\epsilon$ of Δ.

Why the sign is forced. Differentiating the θ's as in (10.8.5), and using the Dirac equation, (10.27.4) and $(\gamma^0)^2=1$,

$$(-i\,\not\partial_x+m)\,\langle T\Psi(x)\bar\Psi(y)\rangle=-i\gamma^0\delta(x^0-y^0)\langle\{\Psi(x),\bar\Psi(y)\}\rangle=-i\delta^4(x-y). \tag{10.27.12}$$

The contact term is the equal-time *anti*commutator because of the relative minus in (10.27.9), so $i\langle T\Psi\bar\Psi\rangle$ is a Green's function. With a $+$ it would be the commutator, whose vacuum value has no γ^0 component at all.

Spacelike separation. D_+ is Lorentz invariant, and a proper orthochronous transformation carries a spacelike ξ to $-\xi$ (go to the frame where $\xi^0 = 0$, rotate by π), so $D_+ = D_-$ there. By (10.27.10) the c-number $\{\Psi(x), \bar\Psi(y)\} = (i\,\partial\!\!\!/ + m)(D_+ - D_-)$ vanishes outside the light cone, and so do the commutators of bilinears such as the charge density. With commutators, $[\Psi(x), \bar\Psi(y)] = (i\,\partial\!\!\!/ + m)(D_+ + D_-)$ would not.

Fermion signs in diagrams

Expanding $e^{i\int \mathcal{L}_1}$ with (10.27.8), every term is a complete contraction. Each pair $\Psi(x)\bar\Psi(y)$ gives $\frac{1}{i}S(x-y)$, and the term carries the sign of the permutation that brings the contracted pairs together in the order $\Psi\bar\Psi$. For free fields the sum over contractions is a determinant,

$$\langle T\Psi(x_1)\bar\Psi(y_1)\cdots\Psi(x_n)\bar\Psi(y_n)\rangle = \det\Big[\frac{1}{i}S(x_k - y_l)\Big]_{k,l=1}^{n}, \tag{10.27.13}$$

spinor indices understood. A vertex $\bar\Psi\Psi$ is even and moves without a sign. The overall sign depends on the order in which the fields are written, which LSZ ties to the order of the creation operators in $|i\rangle$ and $\langle f|$ ($b_1^\dagger b_2^\dagger|0\rangle = -b_2^\dagger b_1^\dagger|0\rangle$); it drops out of $|\mathcal{T}|^2$. Two signs are physical.

Exchange. Two diagrams that differ by swapping the $\bar\Psi$ ends of two lines differ by one transposition, hence by a sign, as the two terms of (10.27.13) at $n = 2$. This is the circled minus of LecNote18 p. 2, between the lines $y_1 \to x_1$, $y_2 \to x_2$ and $y_2 \to x_1$, $y_1 \to x_2$. For $e^-e^- \to e^-e^-$ these are the t- and u-channel diagrams, and the minus is Fermi statistics. In $e^-e^+ \to e^-e^+$ the second term of the same correlator is annihilation, the s-channel (§10.28).

Closed loops. A loop through $w_1, \dots, w_n$ contracts $\Psi(w_k)$ with $\bar\Psi(w_{k+1})$ and $\Psi(w_n)$ with $\bar\Psi(w_1)$. In $\prod_k \bar\Psi(w_k)\Psi(w_k) = \bar\Psi_1(\Psi_1\bar\Psi_2)\cdots(\Psi_{n-1}\bar\Psi_n)\Psi_n$, moving Ψ_n to the front passes $2n-1$ odd fields, so this contraction is

$$-\Big(\frac{1}{i}\Big)^n \operatorname{Tr}\big[S(w_n - w_1)\,S(w_1 - w_2)\cdots S(w_{n-1} - w_n)\big]\,: \tag{10.27.14}$$

a trace and a -1 for every n, including $\langle\bar\Psi\Psi\rangle = -\operatorname{Tr} S(0)/i$. This is the "$-1$ per closed fermion loop" that the note on LecNote19 p. 7 lists among the rules of §45. In the path integral it is the determinant: integrating out Ψ in a fixed background φ gives $Z(\varphi)/Z(0) = \det(D - g\varphi)/\det D = \det(1 - gS\varphi)$, so

$$\ln\frac{Z(\varphi)}{Z(0)} = \operatorname{Tr}\ln(1 - gS\varphi) = -\sum_{n\ge1}\frac{g^n}{n}\operatorname{Tr}\big[(S\varphi)^n\big], \tag{10.27.15}$$

where $S\varphi$ is the kernel $S(x-y)\varphi(y)$ and the trace runs over spinor indices and positions. The nth term is the ring of n vertices: $(ig)^n(\frac{1}{i})^n$, then $(n-1)!$ cyclic contractions against the $1/n!$ of the exponential (the ring's

symmetry factor n, §10.9), and the -1 of (10.27.14). A complex scalar with the coupling $g\varphi\chi^\dagger\chi$ gives $(\det)^{-1}$ instead, and the same rings with $+g^n/n$. These two signs, with the vertex ig, the line $\frac{1}{i}\tilde{S}$ and the external spinors, complete the Feynman rules of §10.28.

SUMMARY

- H contains $-\omega\, dd^\dagger$: only anticommutators, $\{\Psi, \bar{\Psi}\} = \gamma^0\delta^3$, make it bounded below and $Q = N_b - N_d$.
- $d^n\theta = (\det J)^{-1} d^n\theta'$ and $\int e^{\bar{\chi}M\chi} = \det M$; Z_0 has no $\frac{1}{2}$ because Ψ, $\bar{\Psi}$ are independent.
- The sign of T cancels the minus of $\sum_s v_s\bar{v}_s$, so $i\langle T\Psi\bar{\Psi}\rangle = S$, and makes its jump the anticommutator.
- Physical signs: a relative minus between diagrams that exchange identical fermions, and -1 with a trace per closed fermion loop ($\det$, not $1/\det$).